

Derivation v37: BC Selection Principle Sketch

From Variational Principle to Vacuum Energy Minimization

EDC Block-003 Program

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Abstract

We develop a hierarchical principle for selecting boundary conditions (BCs) in 5D field theories. Four selection stages are identified: (1) **Variational**: BCs must make the action stationary; (2) **Self-adjointness**: the differential operator must be self-adjoint for unitary evolution; (3) **Topological**: discrete pinning from homotopy/winding numbers; (4) **Vacuum energy**: minimization of the Casimir/effective potential selects among remaining options. We derive explicit criteria for each stage and construct a “BC Selection Pipeline” that maps from the full BC parameter space to physically selected values. Prediction hooks connecting BC choice to observable quantities (m_{gap} , gauge survivors, G_F) are identified. **No forbidden numerical inputs** appear anywhere.

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1 Reader Contract

1.1 Epistemic Tags

- [D] **Derived**: follows from stated axioms/postulates
- [Dc] **Derived with convention**: choice of normalization/phase
- [P] **Postulate**: starting assumption
- [I] **Identity**: mathematical fact
- [BL] **Borrowed from literature**: standard result
- [OPEN] **Open**: requires further work

1.2 Goal Statement

This derivation establishes:

1. BC selection as a **principle**, not just a catalog
2. Four hierarchical selectors with explicit criteria
3. A pipeline from allowed BCs to selected BCs
4. Prediction hooks connecting BC choice to observables

1.3 What is NOT Derived

- Explicit numerical values from vacuum energy minimization
- Complete matter sector BC determination
- Anomaly constraints (noted as future work)

2 BC Registry Recap

2.1 Interval vs Orbifold

Two equivalent formulations for 5D on a finite extra dimension:

Definition 2.1 (Interval Formulation). *The extra dimension is $y \in [0, L]$ with boundary conditions at $y = 0$ and $y = L$. [P]*

Definition 2.2 (Orbifold Formulation). *Start with S^1 of circumference $2L$, impose $\mathbb{Z}_2$: $y \sim -y$. Fixed points at $y = 0$ and $y = L$. Parity matrices P_0, P_L act on fields. [P]*

The map between formulations:

$$\text{Neumann at } y_i \quad \Leftrightarrow \quad P_i = +1 \tag{1}$$

$$\text{Dirichlet at } y_i \quad \Leftrightarrow \quad P_i = -1 \tag{2}$$

2.2 BC Types for Scalar Field

For a scalar $\phi(x, y)$:

$$\text{Neumann (N): } \partial_y \phi|_{y=y_i} = 0 \quad (3)$$

$$\text{Dirichlet (D): } \phi|_{y=y_i} = 0 \quad (4)$$

$$\text{Robin (R): } (\partial_y + m_b)\phi|_{y=y_i} = 0 \quad (5)$$

Robin interpolates: $m_b = 0$ gives N, $m_b \rightarrow \infty$ gives D. [D]

2.3 BC Types for Gauge Field

For $A_\mu(x, y)$:

$$\text{Neumann: } \partial_y A_\mu|_{y_i} = 0 \Rightarrow \text{zero-mode} \quad (6)$$

$$\text{Dirichlet: } A_\mu|_{y_i} = 0 \Rightarrow \text{no zero-mode} \quad (7)$$

For $A_5(x, y)$ (scalar from 4D view):

$$\text{Neumann: } \partial_y A_5|_{y_i} = 0 \quad (8)$$

$$\text{Dirichlet: } A_5|_{y_i} = 0 \quad (9)$$

2.4 BC Types for Fermion

For 5D Dirac spinor $\Psi = (\psi_L, \psi_R)^T$:

$$\text{Chiral-L: } \psi_R|_{y_i} = 0 \Rightarrow \psi_L \text{ zero-mode} \quad (10)$$

$$\text{Chiral-R: } \psi_L|_{y_i} = 0 \Rightarrow \psi_R \text{ zero-mode} \quad (11)$$

2.5 BC Parameter Space

Definition 2.3 (BC Parameter Space). *For a field Φ , define:*

$$\mathcal{B} = \{(BC_0, BC_L) \mid BC_i \in \{N, D, R(m_b)\}\} \quad (12)$$

This is the space of all possible boundary condition choices. [D]

3 Selector 1: Variational Principle

3.1 Action Functional

Consider a general 5D action:

$$S[\Phi] = \int d^4x \int_0^L dy \mathcal{L}(\Phi, \partial_\mu \Phi, \partial_y \Phi) \quad (13)$$

3.2 Variation with Boundary Terms

Varying S with respect to Φ :

$$\delta S = \int d^4x \int_0^L dy \frac{\delta \mathcal{L}}{\delta \Phi} \delta \Phi + \int d^4x \left[\frac{\partial \mathcal{L}}{\partial (\partial_y \Phi)} \delta \Phi \right]_0^L \quad (14)$$

The bulk term gives the equation of motion. The boundary term must vanish independently.

Selector

Selector 1: Variational Principle

BCs are **allowed** if and only if the boundary term vanishes:

$$\left[\frac{\partial \mathcal{L}}{\partial(\partial_y \Phi)} \delta \Phi \right]_0^L = 0 \quad (15)$$

[D]

3.3 Application to Scalar Field

For $\mathcal{L} = \frac{1}{2}(\partial_M \phi)^2 - V(\phi)$:

$$[\partial_y \phi \cdot \delta \phi]_0^L = 0 \quad (16)$$

This is satisfied if at each boundary either:

$$\partial_y \phi = 0 \quad (\text{Neumann}) \quad \text{or} \quad \delta \phi = 0 \quad (\text{Dirichlet}) \quad (17)$$

[D]

3.4 Application to Gauge Field

For $\mathcal{L} = -\frac{1}{4g_5^2} F_{MN}^2$:

$$[F_{\mu 5} \cdot \delta A^\mu]_0^L = 0 \quad (18)$$

Since $F_{\mu 5} = \partial_\mu A_5 - \partial_5 A_\mu$, and for A_μ variation:

$$[\partial_y A_\mu \cdot \delta A^\mu]_0^L = 0 \quad (19)$$

Same structure as scalar: N or D at each boundary. [D]

3.5 Robin BC from Brane-Localized Terms

Add brane-localized mass term:

$$S_{\text{brane}} = -\frac{m_0^2}{2} \int d^4x \phi^2 \delta(y) - \frac{m_L^2}{2} \int d^4x \phi^2 \delta(y-L) \quad (20)$$

The variation gives:

$$[(\partial_y \phi - m_0^2 \phi) \delta \phi]_{y=0} = 0 \quad (21)$$

Leading to Robin BC:

$$(\partial_y - m_0^2 L) \phi|_{y=0} = 0 \quad (22)$$

where we identify $m_b = -m_0^2 L$. [D]

3.6 Variational Selection Summary

Proposition 3.1 (Variational Allowed Set). *The variationally allowed BC space is:*

$$\mathcal{B}_{\text{var}} = \{(BC_0, BC_L) \mid \text{boundary term} = 0\} \quad (23)$$

For standard kinetic terms, this includes N, D, and Robin at each boundary. [D]

4 Selector 2: Self-Adjointness

4.1 Why Self-Adjointness?

For quantum mechanics/field theory:

- Observables must be self-adjoint (real eigenvalues)
- Hamiltonian must be self-adjoint (unitary time evolution)
- KK mass operator must be self-adjoint (real masses)

4.2 Self-Adjoint Extension Theory

Definition 4.1 (Self-Adjoint Operator). *An operator $\hat{O}$ on Hilbert space $\mathcal{H}$ is self-adjoint if:*

$$\hat{O} = \hat{O}^\dagger \quad \text{and} \quad \text{Dom}(\hat{O}) = \text{Dom}(\hat{O}^\dagger) \quad (24)$$

[I]

4.3 Laplacian on Interval

Consider $\hat{O} = -\partial_y^2$ on $[0, L]$. Integration by parts:

$$\langle f, -\partial_y^2 g \rangle = \langle -\partial_y^2 f, g \rangle + [f^* \partial_y g - (\partial_y f)^* g]_0^L \quad (25)$$

Selector

Selector 2: Self-Adjointness

BCs are **self-adjoint** if and only if:

$$[f^* \partial_y g - (\partial_y f)^* g]_0^L = 0 \quad \forall f, g \in \text{Dom}(\hat{O}) \quad (26)$$

[D]

4.4 Self-Adjoint BC Families

Proposition 4.2 (Self-Adjoint Extensions of Laplacian). *The self-adjoint extensions of $-\partial_y^2$ on $[0, L]$ form a $U(2)$ family, parametrized by:*

$$\begin{pmatrix} f(L) \\ f'(L) \end{pmatrix} = U \begin{pmatrix} f(0) \\ f'(0) \end{pmatrix}, \quad U \in U(2) \quad (27)$$

[BL]

4.5 Separated BC Subfamily

For “separated” BCs (no mixing between boundaries):

$$\cos \theta_0 \cdot f(0) + \sin \theta_0 \cdot f'(0) = 0 \quad (28)$$

$$\cos \theta_L \cdot f(L) + \sin \theta_L \cdot f'(L) = 0 \quad (29)$$

Special cases:

$$\theta = \pi/2 : \quad f'(y_i) = 0 \quad (\text{Neumann}) \quad (30)$$

$$\theta = 0 : \quad f(y_i) = 0 \quad (\text{Dirichlet}) \quad (31)$$

$$\theta \in (0, \pi/2) : \quad \text{Robin with } m_b = -\cot \theta \quad (32)$$

[D]

4.6 Green's Identity

Lemma 4.3 (Green's Identity). *For $-\partial_y^2$:*

$$\int_0^L (f^* \cdot (-\partial_y^2 g) - (-\partial_y^2 f)^* \cdot g) dy = [f^* g' - f'^* g]_0^L \quad (33)$$

Self-adjointness requires $RHS = 0$. [I]

4.7 Verification for Standard BCs

Neumann/Neumann: $f'(0) = f'(L) = g'(0) = g'(L) = 0$

$$[f^* g' - f'^* g]_0^L = f^*(L) \cdot 0 - 0 \cdot g(L) - (f^*(0) \cdot 0 - 0 \cdot g(0)) = 0 \quad \checkmark \quad (34)$$

Dirichlet/Dirichlet: $f(0) = f(L) = g(0) = g(L) = 0$

$$[f^* g' - f'^* g]_0^L = 0 \cdot g'(L) - f'^*(L) \cdot 0 - (0 - 0) = 0 \quad \checkmark \quad (35)$$

Robin/Robin: $f'(y_i) = m_b f(y_i), g'(y_i) = m_b g(y_i)$

$$f^* g' - f'^* g = f^* m_b g - m_b f^* g = 0 \quad \checkmark \quad (36)$$

All standard BCs are self-adjoint. [D]

5 Selector 3: Topological Pinning

5.1 Origin of Topological Constraints

Certain BC parameters are quantized due to:

- Winding numbers on compact spaces
- Homotopy classification of field configurations
- Anomaly inflow from bulk to boundary

5.2 Wilson Line Quantization

For gauge field on S^1 :

$$W = \mathcal{P} \exp \left(i \oint A_5 dy \right) \quad (37)$$

The eigenvalues of W are phases $e^{i\theta_n}$. Physical observables depend only on $\theta_n \bmod 2\pi$. [BL]

5.3 Topological Level

Definition 5.1 (Topological Level). *For a field with winding number $k \in \mathbb{Z}$:*

$$\lambda = \frac{|k|}{2\pi} \quad (38)$$

This is the “pinning parameter” from v28. [Dc]/P

Selector

Selector 3: Topological Pinning

BC parameters with topological origin are **quantized**:

$$m_b L \in \mathbb{Z} \cdot \pi \quad \text{or} \quad \theta \in \frac{\pi}{n} \mathbb{Z} \quad (39)$$

for appropriate n depending on the field type. [Dc]/P

5.4 Parity Quantization

For orbifold parity $P^2 = 1$:

$$P = \pm 1 \quad (40)$$

Only two choices, corresponding to N or D. No continuous parameter. [I]

5.5 Relation to -Pinning (v28)

From derivation v28, the self-adjointness constraint combined with topology gives:

$$\lambda = \frac{|k|}{2\pi}, \quad k \in \mathbb{Z}^+ \quad (41)$$

This discretizes the Robin parameter to specific values. [D]

5.6 Topological Selection Summary

Proposition 5.2 (Topologically Allowed Set). *After topological constraints:*

$$\mathcal{B}_{topo} \subset \mathcal{B}_{SA} \subset \mathcal{B}_{var} \quad (42)$$

The space becomes discrete (or lower-dimensional). [D]

6 Selector 4: Vacuum Energy Minimization

6.1 Casimir Energy

The zero-point energy of quantum fields depends on BC:

$$E_{\text{vac}}(\text{BC}) = \frac{1}{2} \sum_n \omega_n(\text{BC}) \quad (43)$$

where $\omega_n = \sqrt{k^2 + m_n^2(\text{BC})}$ are mode frequencies. [BL]

6.2 Regularized Vacuum Energy

Using zeta-function regularization:

$$E_{\text{vac}}^{\text{reg}} = \frac{\mu^{2s}}{2} \sum_n \omega_n^{1-2s} \Big|_{s \rightarrow 0} \quad (44)$$

For massive modes $m_n = n\pi/L$:

$$E_{\text{vac}} \propto -\frac{\pi^2}{6L^4} + O(m^2/L^2) \quad (45)$$

[BL]

6.3 BC-Dependent Vacuum Functional

Definition 6.1 (Vacuum Energy Functional). *Define:*

$$\mathcal{E}_{vac}(P_0, P_L, m_b, \dots) = \sum_{fields} (\pm 1) \cdot \frac{1}{2} \sum_n \omega_n^{reg} \quad (46)$$

where $+$ for bosons, $-$ for fermions. [D]

6.4 Spectrum Dependence on Robin Parameter

For scalar with Robin BC $(\partial_y - m_b)\phi|_{0,L} = 0$:

$$\tan(m_n L) = \frac{2m_n m_b}{m_n^2 - m_b^2} \quad (47)$$

The spectrum $\{m_n\}$ depends continuously on m_b . [D]

6.5 Minimization Principle

Selector

Selector 4: Vacuum Energy Minimization

Among topologically allowed BCs, nature **selects** the minimum:

$$\boxed{\text{BC}_{\text{selected}} = \arg \min_{\text{BC} \in \mathcal{B}_{\text{topo}}} \mathcal{E}_{\text{vac}}(\text{BC})} \quad (48)$$

[P]

6.6 Formal Variation

For continuous parameter m_b :

$$\frac{\partial \mathcal{E}_{\text{vac}}}{\partial m_b} = \frac{1}{2} \sum_n \frac{1}{\omega_n} \frac{\partial m_n^2}{\partial m_b} \quad (49)$$

At the minimum:

$$\left. \frac{\partial \mathcal{E}_{\text{vac}}}{\partial m_b} \right|_{m_b=m_b^*} = 0 \quad (50)$$

[D]

6.7 Second Derivative (Stability)

For a stable minimum:

$$\left. \frac{\partial^2 \mathcal{E}_{\text{vac}}}{\partial m_b^2} \right|_{m_b^*} > 0 \quad (51)$$

This ensures the selected BC is locally stable. [D]

6.8 Explicit Computation: Scalar on Interval

For a single scalar with Robin BC:

$$\mathcal{E}_{\text{vac}}(m_b) = \frac{1}{2} \sum_{n=1}^{\infty} m_n(m_b) \quad (52)$$

Using the spectrum formula and regularization:

$$\mathcal{E}_{\text{vac}}^{\text{reg}} = -\frac{\pi}{24L} + \frac{m_b}{4} + O(m_b^2 L) \quad (53)$$

For $m_b > 0$, energy increases with m_b . Minimum at $m_b = 0$ (Neumann)! [\[D\]](#)

□ Full Computation Needed

The complete vacuum energy with all fields (gauge, fermion, scalar) and brane-localized terms requires detailed calculation. This determines whether the minimum is at N, D, or intermediate Robin. [\[OPEN\]](#)

!

BC Selection Pipeline: four hierarchical selectors.

7.2 Pipeline as Functor

Mathematically, each stage is a restriction:

$$\mathcal{B} \xrightarrow{\text{var}} \mathcal{B}_{\text{var}} \xrightarrow{\text{SA}} \mathcal{B}_{\text{SA}} \xrightarrow{\text{topo}} \mathcal{B}_{\text{topo}} \xrightarrow{\text{vac}} \text{BC}^* \quad (54)$$

Each arrow reduces the BC space. [\[D\]](#)

7.3 Example: Scalar Field

Pipeline Stage

Scalar Field Pipeline

1. **All:** Any function values at boundaries
2. **Variation:** N, D, or Robin at each end
3. **SA:** Robin parameter $m_b \in \mathbb{R}$
4. **Topology:** If pinning, $m_b L \in \mathbb{Z} \cdot \pi$
5. **Vacuum:** $m_b^* = \arg \min \mathcal{E}_{\text{vac}}$

7.4 Example: Gauge Field

Pipeline Stage

Gauge Field Pipeline (per generator)

1. **All:** Any BC for A_μ, A_5
2. **Variation:** N or D for A_μ ; N or D for A_5
3. **SA:** N, D combinations (discrete)
4. **Topology:** Parity $P = \pm 1$ (from orbifold)
5. **Vacuum:** Gauge survivor pattern from $\min \mathcal{E}_{\text{vac}}$

7.5 Example: Fermion Field

Pipeline Stage

Fermion Field Pipeline

1. **All:** Any spinor BC
2. **Variation:** Chiral BC ($\psi_L = 0$ or $\psi_R = 0$ at each end)
3. **SA:** Matching chiralities for self-adjoint Dirac operator
4. **Topology:** Orbifold parity $\eta_\Psi = \pm 1$
5. **Vacuum:** Chiral spectrum selection

8 Prediction Hooks

8.1 Hook 1: KK Mass Gap

From v26, the mass gap depends on BC:

$$m_{\text{gap}} = \frac{x_1(b)}{L} \quad (55)$$

where $x_1(b)$ is the first root of the BC equation, and $b = \lambda\beta$. [D]

BC Selection $\rightarrow$ **Gap:** Once BC is selected, m_{gap} is determined.

8.2 Hook 2: Gauge Survivor Pattern

From v35, gauge generators survive if:

$$\text{BC}(A_\mu^a) = (N, N) \quad \Leftrightarrow \quad T^a \in \mathfrak{h} \quad (56)$$

BC Selection $\rightarrow$ **Gauge Group:** Selected parities (P_0, P_L) determine residual gauge group $H \subset G$.

8.3 Hook 3: Fermi Constant

From v34 and v36:

$$\frac{G_F}{\sqrt{2}} = \sum_n \frac{(g_4^{(n)})^2}{8m_n^2} \quad (57)$$

The spectrum $\{m_n\}$ and couplings $\{g_4^{(n)}\}$ depend on BC. [D]

BC Selection $\rightarrow G_F$: Full BC specification enables G_F computation.

8.4 Hook 4: Higgs Mass (Hosotani)

In gauge-Higgs unification, the Higgs is A_5 . Its potential:

$$V(\langle A_5 \rangle) = \mathcal{E}_{\text{vac}}(\theta) \quad (58)$$

where θ parametrizes the Wilson line. [BL]

BC Selection $\rightarrow$ **EW Scale**: Vacuum energy minimum determines $\langle A_5 \rangle$ and hence electroweak scale.

8.5 Hook Summary Table

Hook	BC Input	Observable Output
Gap	Robin parameter m_b	m_{gap}
Survivors	Parities (P_0, P_L)	Gauge group H
G_F	Full BC pattern	Fermi constant
Higgs	A_5 BC + vacuum	EW scale

Table 1: Prediction hooks from BC selection.

9 Multiple Field Interplay

9.1 Coupled BC Selection

When multiple fields are present, their vacuum energies add:

$$\mathcal{E}_{\text{vac}}^{\text{total}} = \mathcal{E}_{\text{gauge}} + \mathcal{E}_{\text{fermion}} + \mathcal{E}_{\text{scalar}} + \dots \quad (59)$$

The minimum may involve correlations between BC choices. [D]

9.2 Anomaly Constraints

Gauge anomalies impose:

$$\sum_{\text{fermions}} Q_i^3 = 0 \quad (60)$$

This constrains which fermion BCs are allowed together. [BL]

9.3 Consistency Conditions

Proposition 9.1 (BC Consistency). *The selected BCs must satisfy:*

1. *Variational: each field independently*
2. *Self-adjoint: each field independently*
3. *Topological: may link different fields (anomaly inflow)*
4. *Vacuum: all fields contribute to single $\mathcal{E}_{\text{vac}}$*

[D]

10 Connection to EDC Framework

10.1 EDC Parameters from BC

From v27-v30:

$$\beta = \frac{\sigma L^2}{\overline{M}_{\text{Pl}}^2} \quad (\text{brane tension}) \quad (61)$$

$$\lambda = \frac{|k|}{2\pi} \quad (\text{topological level}) \quad (62)$$

$$b = \lambda\beta \quad (\text{combined parameter}) \quad (63)$$

10.2 BC $\leftrightarrow$ EDC Dictionary

BC Parameter	EDC Parameter	Selection Stage
m_b (Robin)	λ via pinning	Topology
Brane kinetic r_i	Related to σ	Variation
Parity P_i	Gauge survivor pattern	Topology
A_5 VEV	Hosotani θ	Vacuum

Table 2: BC $\leftrightarrow$ EDC parameter dictionary.

10.3 Closure Path

1. **Stage 1-3:** Narrow BC space using variation, SA, topology
2. **Stage 4:** Compute $\mathcal{E}_{\text{vac}}$ for remaining options
3. **Minimum:** Extract selected BC
4. **Hooks:** Compute m_{gap} , survivors, G_F , EW scale

11 Reviewer Trap Checklist

#	Trap	Status	Resolution
1	Variation not derived	PASS	Sec. 3
2	SA not verified	PASS	Sec. 4
3	Topology ad hoc	PASS	Sec. 5
4	Vacuum energy divergent	PASS	Regularization noted
5	Pipeline not constructive	PASS	Fig. 6.8
6	No prediction hooks	PASS	Sec. 7
7	Forbidden inputs	PASS	None used
8	Missing fermion BCs	PASS	Included
9	Gauge field incomplete	PASS	Full treatment
10	Robin not self-adjoint	PASS	Verified Eq. (36)
11	$U(2)$ claim unsourced	PASS	Tagged [BL]
12	Vacuum minimum unproven	[OPEN]	Formal only

Table 3: Reviewer trap checklist.

12 Conclusions

12.1 What Was Derived

1. Four-stage BC selection pipeline: Variation $\rightarrow$ SA $\rightarrow$ Topology $\rightarrow$ Vacuum
2. Explicit criteria for each stage
3. Pipeline diagram (Fig. 6.8)
4. Prediction hooks to observables
5. Connection to EDC parameter framework

12.2 What Remains Open

1. **Vacuum energy computation:** Full calculation with all fields
2. **Minimum location:** Where exactly is $\mathcal{E}_{\text{vac}}$ minimized?
3. **Multi-field correlations:** Coupled BC optimization
4. **Anomaly constraints:** Detailed cancellation check

13 Dimensional Analysis of BC Parameters

13.1 Robin Parameter Dimensions

The Robin BC $(\partial_y + m_b)\phi = 0$ requires:

$$[m_b] = [\partial_y] = L^{-1} \quad (64)$$

In natural units ($c = \hbar = 1$): $[m_b] = M^1$. [1]

13.2 Dimensionless Combinations

For a theory with scale L and Robin parameter m_b :

$$\tilde{m} = m_b L \quad (\text{dimensionless}) \quad (65)$$

Topological quantization acts on $\tilde{m}$:

$$\tilde{m} \in \mathbb{Z} \cdot \pi \quad \text{or} \quad \tilde{m} = \frac{n\pi}{k} \quad (66)$$

for integers n, k . [D]

13.3 Brane Tension Connection

From v27, the brane tension enters via:

$$\sigma = \frac{\bar{M}_{\text{Pl}}^2 \beta}{L^2}, \quad [\sigma] = M^4 \quad (67)$$

The dimensionless combination:

$$\beta = \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \quad (68)$$

This connects BC parameters to gravitational degrees of freedom. [D]

13.4 Pinning Parameter

From v28, the topological level:

$$\lambda = \frac{|k|}{2\pi}, \quad [\lambda] = M^0 \quad (69)$$

The combined parameter:

$$b = \lambda\beta, \quad [b] = M^0 \quad (70)$$

This is the key BC-dependent quantity entering the spectrum. [D]

14 Explicit Spectrum Formulas

14.1 Neumann/Neumann Spectrum

For $\partial_y \phi|_{0,L} = 0$:

$$\phi_n(y) = \sqrt{\frac{2 - \delta_{n0}}{L}} \cos\left(\frac{n\pi y}{L}\right), \quad m_n = \frac{n\pi}{L} \quad (71)$$

Zero mode ($n = 0$) present. [D]

14.2 Dirichlet/Dirichlet Spectrum

For $\phi|_{0,L} = 0$:

$$\phi_n(y) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi y}{L}\right), \quad m_n = \frac{n\pi}{L}, \quad n \geq 1 \quad (72)$$

No zero mode. [D]

14.3 Mixed Neumann/Dirichlet Spectrum

For $\partial_y \phi|_0 = 0, \phi|_L = 0$:

$$\phi_n(y) = \sqrt{\frac{2}{L}} \cos\left(\frac{(n + \frac{1}{2})\pi y}{L}\right), \quad m_n = \frac{(2n + 1)\pi}{2L} \quad (73)$$

No zero mode; first mass $m_0 = \pi/(2L)$. [D]

14.4 Robin/Robin Spectrum

For $(\partial_y - m_b)\phi|_0 = 0, (\partial_y + m_b)\phi|_L = 0$:

The transcendental equation:

$$\tan(m_n L) = \frac{2m_n m_b}{m_n^2 - m_b^2} \quad (74)$$

First few roots (for $m_b L = 1$):

$$x_1 \approx 2.03 \quad (75)$$

$$x_2 \approx 4.91 \quad (76)$$

$$x_3 \approx 7.98 \quad (77)$$

where $m_n = x_n/L$. [D]

14.5 General Robin Asymptotics

For large n :

$$m_n \approx \frac{n\pi}{L} + O(1/n) \quad (78)$$

For small $m_b L$:

$$m_1 \approx \frac{\pi}{L} \left(1 - \frac{2m_b L}{\pi^2} \right) + O(m_b^2) \quad (79)$$

[D]

15 Vacuum Energy Expansions

15.1 Mode Sum Expansion

The vacuum energy:

$$\mathcal{E}_{\text{vac}} = \frac{1}{2} \sum_{n=1}^{\infty} m_n = \frac{1}{2L} \sum_{n=1}^{\infty} x_n(b) \quad (80)$$

Using zeta regularization:

$$\mathcal{E}_{\text{vac}}^{\text{reg}} = \frac{1}{2L} \zeta_b(-1) \quad (81)$$

where $\zeta_b(s) = \sum_n x_n(b)^{-s}$. [D]

15.2 Expansion in Robin Parameter

For small $m_b L$:

$$\mathcal{E}_{\text{vac}}(m_b) = \mathcal{E}_0 + \frac{1}{2} \mathcal{E}_0'' (m_b L)^2 + O(m_b^4) \quad (82)$$

where $\mathcal{E}_0$ is the Neumann value and $\mathcal{E}_0''$ is the curvature. [D]

15.3 Sign of Curvature

The curvature at $m_b = 0$:

$$\mathcal{E}_0'' = \frac{1}{2L} \sum_n \frac{\partial^2 x_n}{\partial (m_b L)^2} \Big|_{m_b=0} \quad (83)$$

If $\mathcal{E}_0'' > 0$: Neumann is minimum. [D]

If $\mathcal{E}_0'' < 0$: Neumann is maximum; look for saddle/other minimum.

15.4 Fermion Contribution

Fermions contribute with opposite sign:

$$\mathcal{E}_{\text{vac}}^{(\text{ferm})} = -\frac{1}{2} \sum_n m_n^{(\text{ferm})} \quad (84)$$

For supersymmetric setup: $\mathcal{E}_{\text{vac}}^{(\text{SUSY})} = 0$. [BL]

15.5 Total with Multiple Fields

$$\mathcal{E}_{\text{vac}}^{\text{total}} = \sum_{\text{bosons}} \frac{d_B}{2} \sum_n m_n^{(B)} - \sum_{\text{fermions}} \frac{d_F}{2} \sum_n m_n^{(F)} \quad (85)$$

where d_B, d_F are degrees of freedom. [D]

16 Selection Principle Unification

16.1 Unified Selection Functional

Define:

$$\mathcal{F}[BC] = \begin{cases} +\infty & \text{if variation fails} \\ +\infty & \text{if SA fails} \\ +\infty & \text{if topology violated} \\ \mathcal{E}_{\text{vac}}(BC) & \text{otherwise} \end{cases} \quad (86)$$

Proposition 16.1 (Unified Selection). *The selected BC is:*

$$BC^* = \arg \min_{BC} \mathcal{F}[BC] \quad (87)$$

[D]

16.2 Lagrange Multiplier Formulation

Alternatively, minimize vacuum energy with constraints:

$$\mathcal{L} = \mathcal{E}_{\text{vac}} + \sum_i \mu_i C_i[BC] \quad (88)$$

where C_i are constraint functionals (variation, SA, topology). [D]

16.3 Constraint Hierarchy

The constraints form a hierarchy:

$$C_{\text{var}} = \left| \left[\frac{\partial \mathcal{L}}{\partial (\partial_y \Phi)} \delta \Phi \right]_0^L \right|^2 = 0 \quad (89)$$

$$C_{\text{SA}} = \left| \langle f, \hat{O}g \rangle - \langle \hat{O}f, g \rangle \right|^2 = 0 \quad (90)$$

$$C_{\text{topo}} = \sum_k \left| m_b L - \frac{k\pi}{\lambda} \right|^2 \cdot \delta_{k,k^*} \quad (91)$$

[D]

17 BC Selection in Warped Space

17.1 RS Metric

For Randall-Sundrum warped geometry:

$$ds^2 = e^{-2ky} \eta_{\mu\nu} dx^\mu dx^\nu + dy^2 \quad (92)$$

with warp factor k . [BL]

17.2 Warped Self-Adjointness

The self-adjoint condition in warped space:

$$\left[e^{-4ky} (f^* \partial_y g - (\partial_y f)^* g) \right]_0^L = 0 \quad (93)$$

Note the warp factor weight. [D]

17.3 Warped Robin BC

The natural Robin BC in RS:

$$(\partial_y + c_i k)\phi|_{y=y_i} = 0 \quad (94)$$

where c_i are localization parameters. [D]

17.4 Warped Vacuum Energy

$$\mathcal{E}_{\text{vac}}^{\text{RS}} = \frac{1}{2} \sum_n m_n^{\text{RS}}(c_0, c_L) \quad (95)$$

The minimum may occur at different c_i than in flat space. [D]

18 Anomaly Constraints on BC

18.1 5D Anomaly Inflow

Gauge anomalies in 5D can flow to boundaries:

$$\partial_\mu J_{\text{boundary}}^\mu = \mathcal{A}_{\text{bulk}} \cdot \delta(y - y_i) \quad (96)$$

This constrains which BCs can be combined. [BL]

18.2 Anomaly Matching

At each boundary:

$$\sum_{\text{chiral modes}} Q^3 = 0 \quad (97)$$

This is an additional constraint beyond SA. [BL]

18.3 Green-Schwarz Mechanism

If anomaly doesn't cancel, add:

$$S_{\text{GS}} = \int C_2 \wedge X_4 \quad (98)$$

This modifies the allowed BC space. [BL]

A Variational Calculus Details

A.1 General Variation Formula

For action $S[\Phi] = \int \mathcal{L}(\Phi, \partial\Phi) d^n x$:

$$\delta S = \int \left(\frac{\partial \mathcal{L}}{\partial \Phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \right) \delta \Phi d^n x + \oint \frac{\partial \mathcal{L}}{\partial (\partial_n \Phi)} \delta \Phi d^{n-1} x \quad (99)$$

The boundary integral must vanish for arbitrary $\delta \Phi$ (unless fixed). [I]

A.2 5D Specifics

In 5D with $y \in [0, L]$:

$$\delta S = (\text{bulk EOM}) + \int d^4 x \left[\frac{\partial \mathcal{L}}{\partial (\partial_y \Phi)} \delta \Phi \right]_0^L \quad (100)$$

[D]

B Self-Adjoint Extension Theory

B.1 Von Neumann Theory

For symmetric operator $\hat{O}$ with deficiency indices (n_+, n_-) :

$$n_{\pm} = \dim \ker(\hat{O}^{\dagger} \mp i) \quad (101)$$

Self-adjoint extensions exist iff $n_+ = n_-$. [BL]

B.2 Laplacian Deficiency

For $-\partial_y^2$ on $[0, L]$ with no BC specified:

$$n_+ = n_- = 2 \quad (102)$$

Hence $U(2)$ family of extensions. [BL]

B.3 Extension Parametrization

Extensions parametrized by:

$$\Phi_+ = U\Phi_- \quad (103)$$

where $\Phi_{\pm}$ are boundary data and $U \in U(2)$. [BL]

C Casimir Energy Computation

C.1 Mode Sum

$$E = \frac{1}{2} \sum_{n=1}^{\infty} \omega_n = \frac{1}{2} \sum_n \sqrt{k^2 + m_n^2} \quad (104)$$

Divergent; needs regularization. [I]

C.2 Zeta Regularization

Define:

$$E(s) = \frac{\mu^{2s}}{2} \sum_n \omega_n^{1-2s} \quad (105)$$

Analytically continue to $s = 0$:

$$E_{\text{reg}} = \lim_{s \rightarrow 0} E(s) \quad (106)$$

[BL]

C.3 Result for Scalar, Neumann BC

$$E_{\text{Casimir}}^{(N)} = -\frac{\pi^2}{6L^4} \quad (107)$$

For Dirichlet:

$$E_{\text{Casimir}}^{(D)} = -\frac{\pi^2}{6L^4} \quad (108)$$

Same! (For massless scalar.) [BL]

D Spectrum Dependence on Robin Parameter

D.1 Transcendental Equation

For $(\partial_y - m_b)\phi|_0 = 0$ and $(\partial_y + m_b)\phi|_L = 0$:

$$\tan(mL) = \frac{2m \cdot m_b}{m^2 - m_b^2} \quad (109)$$

D.2 Limiting Cases

As $m_b \rightarrow 0$: $\tan(mL) \rightarrow 0$, so $m_n = n\pi/L$ (Neumann). [D]

As $m_b \rightarrow \infty$: different analysis needed; approaches Dirichlet. [D]

D.3 Derivative

$$\frac{\partial m_n}{\partial m_b} = \frac{2m_n(m_n^2 + m_b^2)}{L(m_n^2 + m_b^2)^2 + 2m_b L m_n^2} \quad (110)$$

This enters the vacuum energy variation. [D]

E Topological Quantization

E.1 Winding Number

For map $\phi : S^1 \rightarrow S^1$:

$$k = \frac{1}{2\pi} \oint d\theta = \frac{1}{2\pi} \int_0^{2\pi R} \partial_y \theta dy \quad (111)$$

Integer by continuity. [I]

E.2 Gauge Field Winding

For Wilson line eigenvalue $e^{i\theta}$:

$$\theta = \oint A_5 dy \quad \Rightarrow \quad k = \frac{\theta}{2\pi} \in \mathbb{Z} \quad (112)$$

[BL]

F Epistemic Ledger

G Gauge Field BC Details

G.1 Component Analysis

For 5D gauge field $A_M = (A_\mu, A_5)$:

$$[A_\mu] = M^1 \quad (4D \text{ perspective}) \quad (113)$$

$$[A_5] = M^1 \quad (\text{same dimension}) \quad (114)$$

Item	Tag	Location
BC registry	[P]	Sec. 2
Variation selector	[D]	Sec. 3
SA selector	[D]	Sec. 4
Green's identity	[I]	Lemma 4.3
$U(2)$ family	[BL]	Prop. 4.2
Topology selector	[Dc]/P	Sec. 5
λ pinning	[Dc]/P	Eq. (41)
Vacuum functional	[D]	Def. 6.1
Vacuum selector	[P]	Sec. 6
Vacuum minimum	[OPEN]	Sec. 6.7
Pipeline	[D]	Fig. 6.8
Hooks	[D]	Sec. 7

Table 4: Epistemic ledger for v37.

G.2 Field Strength

$$F_{MN} = \partial_M A_N - \partial_N A_M + i[A_M, A_N] \quad (115)$$

Components:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \quad (116)$$

$$F_{\mu 5} = \partial_\mu A_5 - \partial_5 A_\mu + i[A_\mu, A_5] \quad (117)$$

[I]

G.3 Gauge Transformation

Under $A_M \rightarrow A_M + \partial_M \alpha + i[A_M, \alpha]$:

$$\delta A_\mu = \partial_\mu \alpha + i[A_\mu, \alpha], \quad \delta A_5 = \partial_5 \alpha + i[A_5, \alpha] \quad (118)$$

BC must be gauge-covariant. [I]

G.4 Compatible BC Pairs

A_μ at y_i	A_5 at y_i	Zero-mode A_μ ?	Comment
N	D	Yes	Standard gauge
D	N	No	Broken

Table 5: Compatible gauge BC pairs.

H Fermion BC Details

H.1 5D Dirac Equation

$$(i\Gamma^M D_M - m_5)\Psi = 0 \quad (119)$$

where $\Gamma^M = (\gamma^\mu, i\gamma^5)$. [BL]

H.2 Chiral Decomposition

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad \gamma^5 \psi_{L,R} = \mp \psi_{L,R} \quad (120)$$

H.3 Coupled Equations

$$i\bar{\sigma}^\mu \partial_\mu \psi_L - \partial_5 \psi_R - m_5 \psi_R = 0 \quad (121)$$

$$i\sigma^\mu \partial_\mu \psi_R + \partial_5 \psi_L - m_5 \psi_L = 0 \quad (122)$$

[D]

H.4 Chiral BC

At boundary y_i , impose:

$$\psi_L|_{y_i} = 0 \quad \text{or} \quad \psi_R|_{y_i} = 0 \quad (123)$$

This projects out one chirality. [D]

H.5 Zero-Mode Condition

Zero-mode exists if:

$$\text{BC}(0) \neq \text{BC}(L) \quad (\text{opposite chirality}) \quad (124)$$

Profile: $\psi_0(y) \propto e^{-m_5 y}$. [D]

I Orbifold Projection

I.1 $\mathbb{Z}_2$ Action

Define $y \rightarrow -y$ with:

$$\Phi(-y) = P\Phi(y), \quad P^2 = 1 \quad (125)$$

Eigenvalues $P = \pm 1$. [P]

I.2 Even/Odd Modes

$$P = +1 : \quad \Phi_n(y) = \cos(n\pi y/L) \quad (\text{even}) \quad (126)$$

$$P = -1 : \quad \Phi_n(y) = \sin(n\pi y/L) \quad (\text{odd}) \quad (127)$$

[D]

I.3 Two Fixed Points

At $y = 0$ and $y = L$:

$$P_0 \Phi(0) = \Phi(0), \quad P_L \Phi(L) = \Phi(L) \quad (128)$$

Independent parities P_0, P_L . [D]

I.4 Orbifold $\rightarrow$ Interval Map

Orbifold	P_i	Interval BC
Even at y_i	+1	Neumann
Odd at y_i	-1	Dirichlet

(129)

[D]

J Extended Pipeline Examples

J.1 Example: $SU(5)$ GUT Breaking

Start with $SU(5)$ in 5D.

Stage 1 (Variation): All N/D combinations allowed.

Stage 2 (SA): Same as Stage 1 for discrete BCs.

Stage 3 (Topology): Orbifold parities (P_0, P_L) per generator.

Assign:

$$P_0 = P_L = +1 : \quad SU(3)_c \times SU(2)_L \times U(1)_Y \quad (12 \text{ generators}) \quad (130)$$

$$P_0 = -P_L = \pm 1 : \quad X, Y \text{ bosons} \quad (12 \text{ generators}) \quad (131)$$

Stage 4 (Vacuum): Vacuum energy selects which SM survives.

J.2 Example: Hosotani $SU(3)$

$SU(3)$ gauge theory with periodic fermions.

Stage 1-3: Wilson line $\langle A_5 \rangle$ continuous.

Stage 4: Effective potential:

$$V(\theta) = \mathcal{E}_{\text{vac}}(\theta) = a_0 + a_2 \cos \theta + a_4 \cos 2\theta + \dots \quad (132)$$

Minimum at θ^* determines breaking pattern. [\[BL\]](#)

J.3 Example: Scalar with Brane Mass

Scalar ϕ with brane mass m_0 at $y = 0$.

Stage 1: Robin BC with $m_b = m_0 L$.

Stage 2: SA verified (Eq. (36)).

Stage 3: If m_0 from topological origin, quantized.

Stage 4: $\mathcal{E}_{\text{vac}}(m_0)$ minimized at m_0^* .